

Review intelligence for fit, usage, and VOC decisions

adiFit converts product reviews into structured signals for fit, comfort, design, use cases, customer segments, and actionable merchandising insights.

WHAT IT BUILDS

리뷰에서 사이즈, 착용감, 색상/디자인, 용도, VOC 이슈를 추출하는 AI 리뷰 분석 도구.

HOW IT WORKS

Gemini structured output → embeddings/RAG →
Streamlit dashboard → clustering & trend view.
Repo:
github.com/HyeokyYun/26ACEProgram_TechDigitalOmni

KEY RESULT

24개 데모 리뷰, 3개 상품, 5개 고객 세그먼트, size signal accuracy 95.8%.

CHALLENGE & SOLUTION

근거 없는 사이즈 추천을 막기 위해 검색된 리뷰 안에서만 답하도록 제한하고, review_id citation과 coverage check를 붙였습니다.

NEXT STEPS

대규모 first-party 리뷰, 반품/CS 데이터, 상품 카탈로그를 연결해 VOC drift를 운영 지표로 확장.

Reviews contain the signal, but the signal is buried.

사이즈, 발볼, 착용감, 디자인, 용도, 품질 이슈가 자유 텍스트 안에 섞여 있어 고객과 내부 담당자 모두 빠르게 판단하기 어렵습니다.

Consumer friction

“정사이즈로 사도 될까?”, “발볼 넓으면 괜찮을까?”, “출퇴근용으로 오래 신어도 편할까?” 같은 질문을 리뷰에서 직접 찾아야 합니다.

Operational friction

핏 불만, 내구성, 배송/포장, 가격 저항 같은 VOC를 사람이 읽고 분류해야 하며, PDP 개선 액션으로 연결되기 어렵습니다.

Data challenge

리뷰는 짧고 비정형이며 한 문장 안에 긍정, 부정, 사용 맥락이 함께 나타납니다.

AI opportunity

LLM으로 리뷰를 구조화하고, 검색/집계/클러스터링을 통해 고객 경험과 운영 의사결정을 동시에 지원합니다.

One analysis engine, two practical interfaces.

FOR CONSUMERS

Grounded Fit Advisor

사용자의 평소 사이즈, 발볼, 질문을 입력하면 관련 리뷰를 검색하고, 근거 인용과 함께 사이즈 조언을 생성합니다.

“평소 260 신고 발볼 넓은데 정사이즈로 사도 될까요?”

FOR TEAMS

Merchandiser VOC Dashboard

상품별 사이즈 신호, 착용감, 디자인/색상, 주요 VOC 이슈, 세그먼트, 월별 트렌드를 시각화하고 PDP 개선 액션으로 연결합니다.

“Samba OG: 발볼/사이즈업 리스크가 반복됩니다.”

A lightweight pipeline designed for a working prototype.

1

Review data

24 curated demo reviews across Ultraboost Light, Samba OG, and Adizero Adios Pro 3.

2

Structured extraction

Gemini extracts size, width, aspect sentiment, use cases, comfort, design/color, customer tags, and VOC issues.

3

Retrieval & clustering

Embeddings support RAG search, KMeans segmentation, PCA projection, and monthly tag aggregation.

4

Decision surfaces

Fit advisor, merchandising dashboard, product comparison, segment/trend view, and validation tab.

Why this shape

The pipeline is lightweight enough to run as a demo while preserving clear boundaries between extraction, retrieval, aggregation, segmentation, and UI.

What scales next

The same contracts can accept live reviews, larger vector indexes, product catalog metadata, and return/CS signals without changing the workflow.

Unstructured reviews become reusable product signals.

The schema keeps the LLM output consistent across retrieval, aggregation, visualization, and evaluation.

SIGNAL	EXAMPLES	USED FOR
Fit & size	runs_small, true_to_size, runs_large, width signal	Advisor, PDP guidance
Aspect sentiment	fit_size, comfort, quality, durability, delivery, design, value	VOC dashboard
Usage context	running, walking, commute, daily, gym, race	Customer segments
Comfort & design	cushioned, responsive, breathable, easy_to_style, gets_dirty	Messaging and product comparison
VOC issues	fit_risk, durability, delivery_packaging, price_resistance	Action recommendations

Answers are generated from retrieved review evidence, not general guesses.

INPUT

Product: Samba OG
Usual size: 260
Width: wide
Question: 정사이즈로 사도 될까요?

RETRIEVAL

Top-k product reviews are retrieved by embedding similarity and passed to the LLM as the only allowed evidence.

OUTPUT

Recommendation, confidence, rationale, caveats, and cited review snippets are displayed in the app.

Recommended response pattern: size up for wide feet, with review_id citations and caveats for fit risk.

Review analysis is translated into merchandising actions.

Product-level insight

리뷰 수, 평균 평점, 사이즈 결론, 발볼 결론, 속성별 감성, 사이즈 신호 분포, 상위 이슈를 한 화면에서 확인합니다.



VOC → action

Rules convert recurring signals into PDP and operations recommendations.

- Show half-size-up guidance when small-fit signals dominate.
- Recommend alternative models for wide-foot risk.
- Surface durability, delivery, or price resistance as operating VOC.

Clustering separates review patterns into usage segments.

Embeddings create clusters; structured tags explain the clusters in business language.

Comfort-driven Buyers

쿠션, 반발력, 장시간 착용 편안함 중심.

Wide-foot Fit Risk

발볼 넓음, 조임, 사이즈업, 교환/환불 위험 중심.

Performance Runners

러닝, 대회, 기록, 반발력, 경량성 중심.

Post-purchase VOC Risk

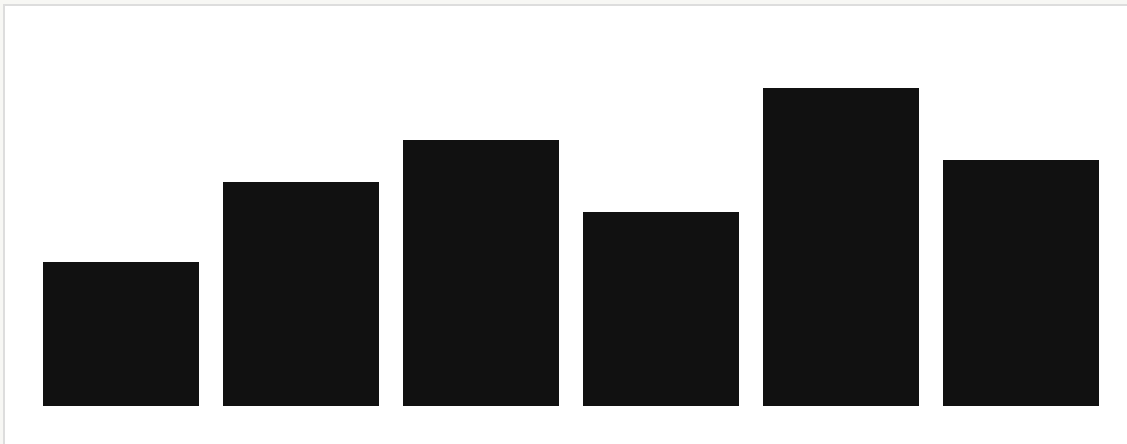
내구성, 배송/포장, 가격 저항 같은 구매 후 이슈 중심.

At larger scale, the same approach can detect customer segment drift and emerging product issues over time.

Monthly VOC tags make issue movement visible.

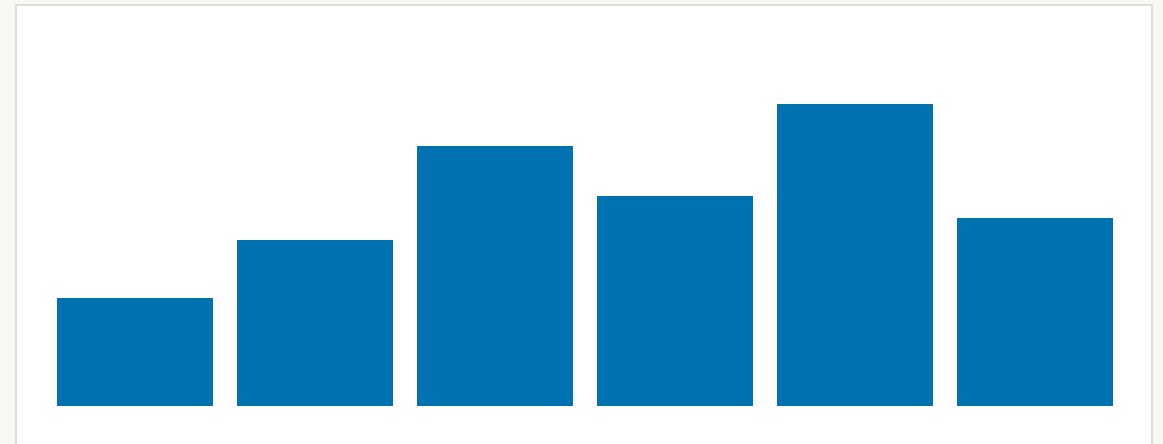
Issue trend

Fit risk, durability, delivery/packaging, price resistance, and comfort complaints are tracked by month.



Use-case trend

Running, daily, commute, race, and long-wear comfort signals can be monitored as seasonal demand patterns.



The prototype includes a lightweight validation layer.

24

REVIEW SAMPLES

3

PRODUCTS

95.8%

SIZE SIGNAL ACCURACY

100%

CITATION COVERAGE TESTS

Extraction validation

Manual gold labels check whether the model correctly extracts size_signal for each review. One ambiguous review is intentionally visible in the validation tab.

Grounding validation

Citation coverage checks whether every cited review_id exists in the retrieved evidence set. This is a grounding check, not a complete factuality benchmark.

The current evaluation is intentionally narrow and transparent: it proves the core loop works, while larger datasets are needed for full production-quality benchmarking.

From prototype to operational review intelligence.

DATA SCALE

Connect first-party review streams, return reasons, CS tickets, product catalog, and PDP analytics.

MODEL QUALITY

Expand evaluation beyond size signal into aspect sentiment, citation faithfulness, and segment stability.

BUSINESS LOOP

Trigger PDP copy updates, fit guidance, alternative recommendations, and VOC alerts from emerging signals.

STRUCTURED REVIEWS

GROUNDED ADVICE

CUSTOMER SEGMENTS

VOC TREND MONITORING